

Semester	Fall 2024
Course Code	AML-2404
Section	Section 1
Project Title	AI-Powered Customer Service Chatbot Using Python and NLTK
Group Name	Dev Pankajkumar Bajaniya, Jatin Vaghela, Pratikkumar Bhupendrabhai Chavda, Amal Satheesan, Karop Dezosa Sebastian
Student Names / IDs	Dev Jatin Pratik Amal Dezosa
Reporting Week	Week 3 — [6-May] to [5-May]
Faculty Supervisor	William Pourmajidi
Team Lead this Week	Karop Dezosa Sebastian

Grade the level of collaboration between team members (to be done by Team Lead):

(use: good, medium, below expectation)

Dev	Jatin	Pratik	Amal	Dezosa
Good	Good	Good	Good	Good

Progress Made in Reporting Week at Individual and Group Level

(Detailed progress for Week 3 — continued from Week 2 milestones)

Individual Progress:

Dev (Backend Developer):

- Connected the Flask backend API to the React frontend interface built in Week 2, enabling end-to-end data flow between the NLP engine and the chat UI.
- Implemented WebSocket support using Flask-SocketIO with eventlet for real-time message rendering without page refresh.
- Tested 10 full conversation flows end-to-end; identified and resolved 3 routing bugs in the API layer.

Jatin (AI / ML Specialist):

- Expanded the training dataset by adding 30 new patterns across 5 underrepresented intent categories (e.g., payment disputes, account recovery).
- Retrained the intent classifier on the updated dataset; overall accuracy improved from 88% (Week 2) to 91%, surpassing the Week 3 target of 90%.
- Tuned the confidence threshold from 0.6 to 0.65 to reduce false positives on ambiguous queries.

Pratik (Database Manager):

- Completed MongoDB schema design for storing conversation logs, user sessions, and intent analytics.
- Integrated the database layer with the Flask backend; verified read/write operations for chat history and session tracking.
- Set up indexing on session ID and timestamp fields to ensure fast query performance at scale.

Amal (Frontend / UI Designer):

- Translated wireframes into a fully functional React chat interface with real-time message rendering via WebSocket.
- Added typing indicator animation and message timestamps to improve user experience.
- Mobile responsiveness is partially complete; layout adjustments for smaller screens are ongoing and will be finalized in Week 4.

Dezosa (Documentation & Deployment):

- Updated all project documentation with Week 3 changes, new module descriptions, and updated setup instructions.
- Reviewed and coordinated team pull requests; resolved 2 merge conflicts between the UI integration and NLP branches.
- Began drafting deployment documentation for Render/Vercel cloud platform as outlined in the project proposal.

Team Progress:

- Successfully completed the UI Integration milestone — the chatbot now runs fully end-to-end in the browser via the React frontend and Flask backend.
- Intent classification accuracy reached 91%, exceeding the Week 3 target and progressing ahead on the AI/NLP Integration phase.
- Multi-turn context tracking implemented, enabling the chatbot to handle follow-up queries within the same session.
- MongoDB database layer is live and logging all chat interactions for future analytics and model improvement.

Areas / Tasks Where Progress Could Not Be Made or Difficulties Encountered

Individual Blockers:**Dev:**

- Flask-SocketIO required additional async configuration (eventlet mode) to run correctly in the development environment; resolved after debugging.

Jatin:

- Sourcing additional training patterns for rare intent categories (e.g., payment disputes) was time-consuming; some categories still need more examples in Week 4.

Pratik:

- Initial MongoDB connection pooling caused intermittent timeouts under concurrent test load; resolved by adjusting pool size and timeout settings.

Amal:

- Mobile-responsive layout for the chat UI is incomplete; the component structure needs refactoring for smaller screen breakpoints — deferred to Week 4.

Dezosa:

- No major blockers this week; minor coordination delays from merge conflict resolution were handled promptly.

Team Blockers:

- End-to-end multi-turn conversation testing revealed a context-drop issue after 3+ follow-up queries; root cause identified (session state not persisting across API calls) and a fix is in progress.
- Deployment packaging was deferred to Week 4 due to time spent completing the UI integration milestone.

Tasks to be Completed Next Week (Week 4) at Individual and Group Level

Individual Tasks:**Dev:**

- Fix the multi-turn session state persistence bug identified in Week 3 testing.
- Begin deployment packaging using Docker; configure Render environment variables for the Flask API.

Jatin:

- Continue expanding training data for low-coverage intent categories; run a final retraining cycle targeting 92%+ accuracy.
- Begin implementing a basic sentiment analysis module as outlined in the future enhancements roadmap.

Pratik:

- Implement conversation analytics queries (most common intents, fallback trigger rate) from MongoDB logs.
- Write unit tests for all database operations and verify data integrity under concurrent load.

Amal:

- Complete mobile-responsive UI and conduct usability testing with at least 5 external test users.
- Refine the chat interface visual design based on feedback from the Week 3 team review.

Dezosa:

- Prepare full deployment documentation and finalize the project report first draft.
- Prepare the demo environment for the final presentation rehearsal.

Team Tasks:

- Run full end-to-end testing covering all 12 intent categories, multi-turn flows, and edge cases (ambiguous input, empty input, very long queries).
- Begin rehearsing the final presentation with a live chatbot demonstration.
- Submit the project report draft to Professor William Pourmajidi for review.

Project Completed Percentage (Estimate): 75% — On Schedule
